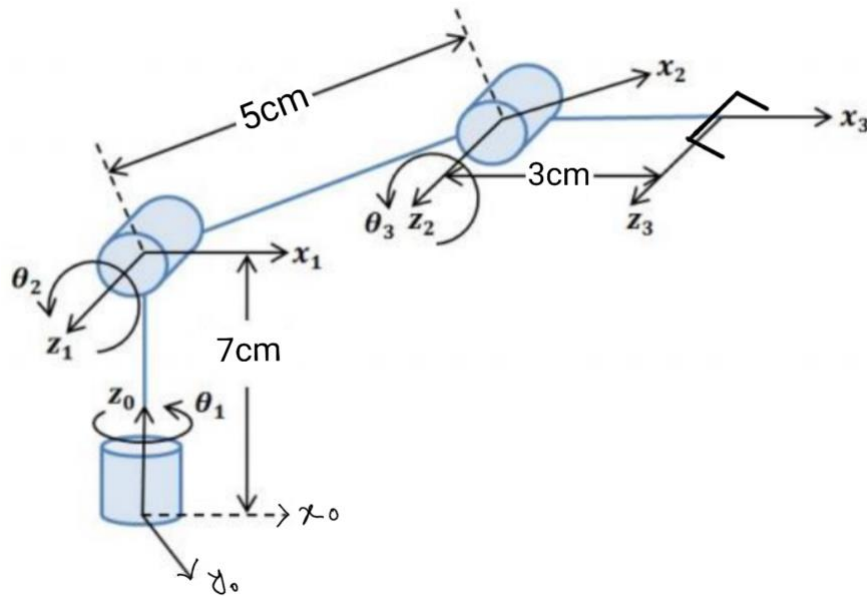




The given manipulator has 3 Degrees of freedom,

- A. Let's assume, the first joint rotated at 47° , the second joint 30.2° and rotation at the third joint was 105° . What should be the position of the end effector after all these joint rotations? [5 marks]
- B. Now, you are required to grab an object by moving the end effector of the given manipulator to the location of the object. If the position of the object is $(X,Y,Z) = (6,8,10)$, what is the amount of rotation needed at each joint? [5 marks]